

Data Modelling - Dementia Group

What is the scope of the database?

The societal problem we selected is the rising rate of dementia in the US. We want to help hospitals prepare for the incoming rise of dementia cases and inform the population about sound habits to decrease the probability of developing dementia later in life.

As such our database aims at supporting US hospitals and health organisations to better study the phenomenon.

How does the data interact?

- A patient lives in exactly one state, but a state can have many patients;
 - A patient can have no or multiple treatments, and treatment can be used on one or many patients;
 - A patient can have no or many types of insurance, and an insurance can cover multiple patients;
 - A patient can have 0 to many GP, and a GP can operate on many patients
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- A treatment can be offered in many states, and a state can have 0 or many treatments
 - A treatment can be covered by zero or many insurance types, and an insurance can cover 0 to many treatment
 - A treatment can be recommended by zero or many GPs, and A GP can prescribe 0 to many treatments
 - A state can have 1 to many GPs, but a GP only operates in 1 state
 - A state can have 1 or many insurance companies offering insurance, and insurance companies can cover 1 or many states;
 - An insurance can cover 0 to many GPs, and a GP can be covered by 0 or many insurances

Entities

- Patient: it represents one patient;
- Treatment: it represents one therapy patients can go through;
- Insurance: it represents one insurance company;
- GP: it represents one general practitioner;

Relationships (DISCUSS Treatment primary key)

- Patient → lives → State (One-to-many relationship);
- Patient → uses → Treatment (Many-to-many relationship);
- Insurance → covers → Patient (Many-to-many relationship);
- GP → operates → Patient (Many-to-many relationship);
- Treatment → offered → State (Many-to-many relationship);
- Insurance → covers → Treatment (Many-to-many relationship);
- GP → prescribes → Treatment (Many-to-many relationship);
- GP → operates → State (One-to-many);
- Insurance → covers → State (Many-to-many);
- Insurance → covers → GP (Many-to-many);

Attributes

Patient

- Patient ID;
- Age;
- Sex;
- Ethnicity;
- Diet;
- BMI: Body Mass Index;
- Income;
- Type of dementia;
- State: state of residence;
- Education: education level;
- Marital Status;
- Smoker: are they smokers?;
- Treatment: type of treatment received to treat dementia;
- Diabetes: have they experienced any type of diabetes?;
- Depression: have they experienced any type of depression?;
- Deafness: do they have hearing deficits?
- Sleep disorders: do they have any sleep disorders?
- Cardiovascular diseases: do they have cardiovascular diseases?

Insurance

- Insurance_company_name: what is the name of the insurance company;
- Insurance_plan: plan offered by the insurance company to the customer;
- Number_of_people_covered: number of people covered by each insurance company;

General Practitioner

- GP_ID: the ID number of the general practitioner;
- Price: a visit cost;
- State: state in which the medical practice that the GP works at is located;
- Medical_practice_name: Name of the practice at which GP works;
- GP_name: First and last name of the GP;

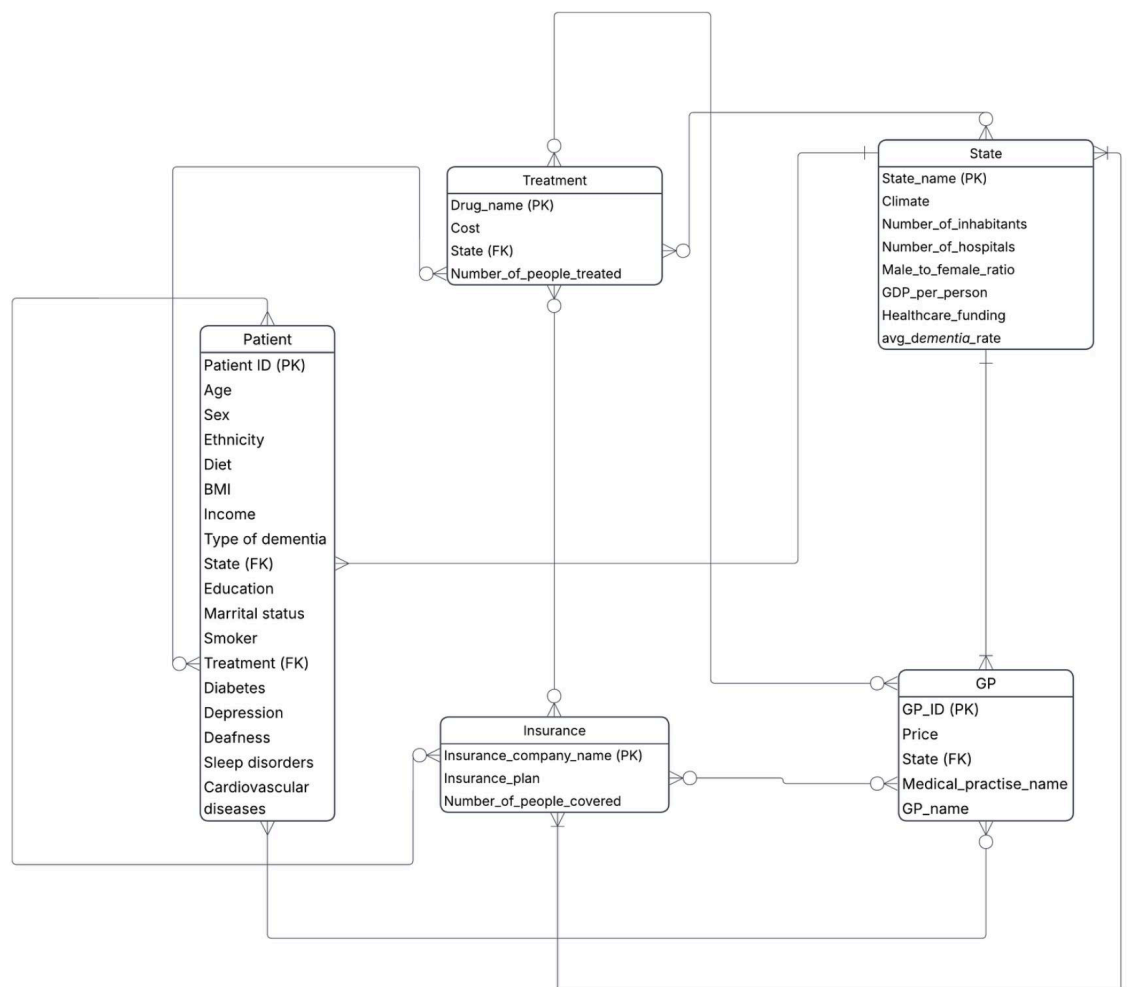
State

- State_name;
- Climate: type of climate of the state;
- Number_of_inhabitants;
- Number_of_hospitals;
- Male_to_female_ratio: how many males are there for every female;
- GDP_per_person: GDP per capita of the state;
- Healthcare_funding: annual amount of healthcare funding received by the federal government;
- Avg_dementia_rate: number of people with dementia in the state, divided by the total number of people in the state;

Treatment

- Drug_name: type of medication used in a treatment;
- Cost: cost of production of the medication;
- State: state in which this medication is being offered;
- Number_of_people_treated: total number of people treated by a medication;

ERD prior to Normalization



Normalization

After creating the bridge tables, which results can be seen in the final ERD diagram below, we move to normalization.

Before and After normalization

1) 1NF normalization

Before:

Patient_ID	Treatment
1	Donepezil, Memantine, Cognitive stimulation therapy, Validation therapy
2	Donepezil, Reminiscence therapy
3	Rivastigmine, Occupational therapy
4	Galantamine

After:

Patient_ID	Treatment
1	Donepezil
1	Memantine
1	Cognitive stimulation therapy
1	Validation therapy
2	Donepezil
2	Reminiscence therapy

3	Rivastigmine
3	Occupational therapy
4	Galantamine

The *after* table, however, was not achieved by modifying any existing entity. Since one patient can have multiple treatments and one treatment can be given to multiple patients, we have a many-to-many connection that violates 1NF but requires a **bridge table** to proceed. Thus the *after* table is a result of applying a bridge table between the entities "Patient" and "Treatment".

Treatment and state were also in violation of 1NF as one treatment may be offered in multiple different states, hence creating a list inside the cell.

Lastly, there was a 1NF violation between the entities "Patient" and "Insurance", such that it was possible for one patient to be subscribed to multiple insurance plans, and one insurance plan to cover multiple patients. This would create the issue of a list in either entity, and as done previously, the issue was resolved by adding a bridge table between the entities.

2) 2NF normalization

Violating 2NF means that a non-key column depends on just a part of the primary key. In our case, since the entities do not possess composite primary keys, we found no 2NF violations.

However, our bridge tables do contain some composite keys. In this case, we found no 2NF problems as well, since all the non-key columns entirely depend on them.

3) 3NF normalization

The cost and insurance coverage in the treatment table were in violation of 3NF as when the cost goes up or down the coverage also changes. We fixed this by creating a new table in which cost is a foreign key and so is therapy and coverage is an attribute of this table instead of the main treatment table.

Before:

Drug_ID (PK)	Cost	Insurance coverage
Donepezil	\$10	100%
Memantine	\$20	10%
Galantamine	\$15	100%

After:

Drug_ID (PK)	Cost (FK)	Insurance coverage
Donepezil	\$10	100%
Memantine	\$20	10%
Galantamine	\$15	100%

ERD After Normalization

